Integrated Retail Analytics for Store Optimisation

Machine Learning & Data Science Capstone Project

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Project Overview

Anomaly Detection

Identify and remove corrupted sales spikes before modelling

Segmentation

Group stores into performance tiers for targeted strategy

Forecasting

Predict weekly demand with high accuracy across all departments

Association Analysis

Uncover cross-selling relationships between departments

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THE BUSINESS PROBLEM

Retailers Are Flying Blind

Inaccurate demand forecasting leads to costly **stockouts** and **dead inventory** — eroding margins and customer trust simultaneously.

External Pressures

Holiday surges, inflation, and fuel costs create unpredictable demand swings

Data Complexity

Sparse historical markdowns and skewed sales distributions distort model training

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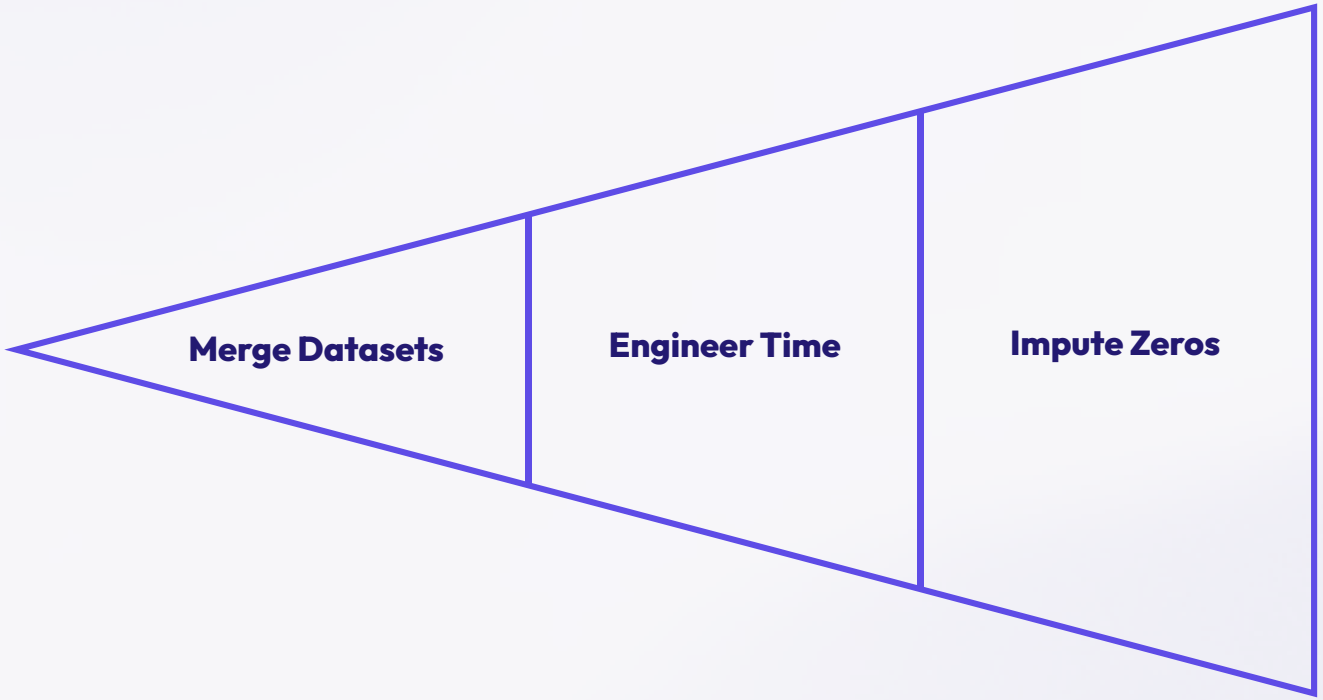
Data & Feature Engineering



A deliberate preprocessing strategy preserved the majority of historical records whilst building richer temporal signals for downstream models.

3 Datasets Merged

Sales, Stores, and Features unified into a single analytical frame

80% Records Retained

Zero-imputation on sparse markdown columns avoided mass row deletion

Temporal Features

Week and Month extracted to capture seasonality and holiday patterns

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Anomaly Detection with Isolation Forest

Extreme, corrupted sales spikes distort model learning. An **unsupervised Isolation Forest** was deployed to identify and remove the top **1%** of outliers before any forecasting took place — ensuring the downstream Random Forest model trained on clean, representative data.

Unsupervised Approach

No labelled anomaly data required

Top 1% Filtered

Extreme spikes isolated and removed

Cleaner Training Data

Model learns from representative patterns

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Demand Forecasting: Random Forest

0.978

R^2 Score

Variance explained by the model

$2,756

RMSE

Average prediction error per week

Key Findings

A **Random Forest Regressor** delivered near-perfect predictive accuracy after anomaly filtering. Feature importance analysis revealed two dominant drivers of weekly sales:

- **Store Size** — the single strongest predictor of sales volume
- **Department** — captures product-mix variation across locations
- Holiday flags and temporal features provided secondary lift

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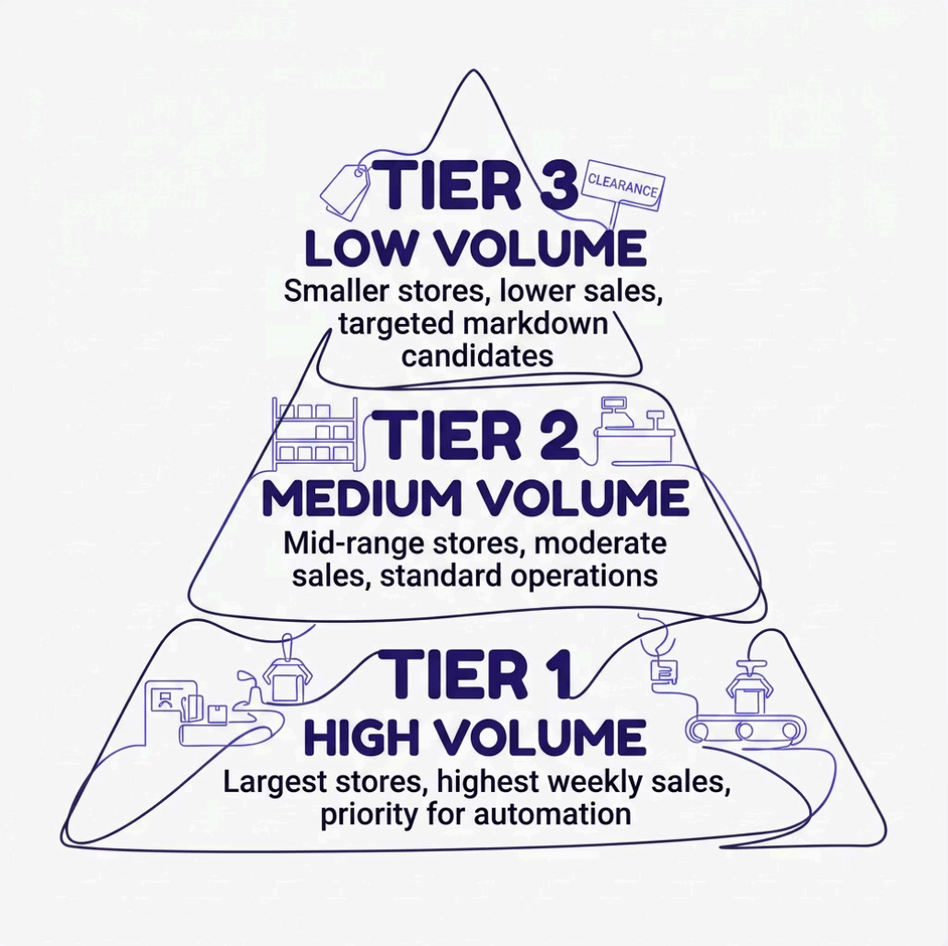
Store Segmentation: K-Means Clustering



Clustering Results

K-Means grouped all **45 stores** into three distinct performance tiers, validated by a **Silhouette Score of 0.58** — indicating well-separated, meaningful clusters.

Tier 1 — High Volume

Largest stores; primary candidates for inventory automation

Tier 2 — Medium Volume

Stable performers with standard operational strategies

Tier 3 — Low Volume

Smaller stores; most responsive to targeted markdowns

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Market Basket & External Factors

Department Cross-Selling

Pearson Correlation analysis revealed strong co-purchase relationships between departments. **Dept 33 & 24** emerged as the top pairing — a prime candidate for joint promotions and co-located merchandising.

Key insight: Co-located high-correlation departments can increase basket size without additional marketing spend.

External Macro Factors

Statistical testing across unemployment rates and fuel price fluctuations demonstrated that neither variable **significantly erodes baseline sales** — suggesting retailer resilience to moderate macroeconomic headwinds.

This finding allows forecasting models to de-prioritise macroeconomic inputs in favour of store-level and temporal features.

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Challenges & Optimisation

Challenge: Skewed Sales Data

Weekly sales distributions were highly right-skewed, with a small number of stores and holidays generating outsized volumes

Challenge: Sparse Markdowns

Historical markdown columns contained extensive gaps, threatening to eliminate the majority of training records if dropped naively

Solution: Tree-Based Models

Random Forest inherently resists skewed distributions — eliminating the need for log transformations and preserving model interpretability

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Strategic Recommendations

Actionable insights derived from all four analytical modules — ready for immediate operational deployment.

Automate Tier 1 Inventory

Deploy automated replenishment systems at high-volume stores to eliminate stockouts during peak demand periods

Bundle Correlated Departments

Design joint promotions for strongly correlated department pairs (e.g., Dept 33 & 24) to increase average basket value

Targeted Markdowns

Direct markdown budgets toward Tier 3 stores during high-inflation periods to defend foot traffic and preserve revenue

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